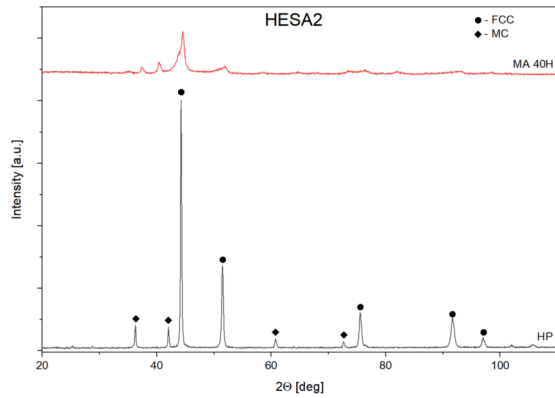
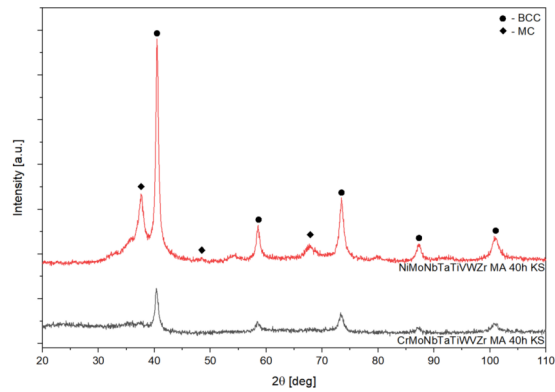


(a) SPARK-S1 (Ni-based HESA): MA powder vs HP compact



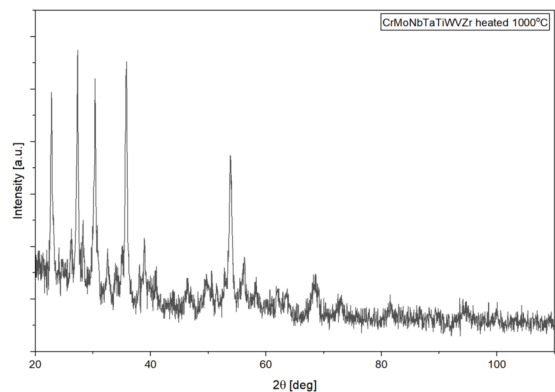
FCC matrix + minor MC carbide phase confirmed in HP compact.

(b) Eight-element MA powders: SPARK-H2 vs SPARK-H3



BCC matrix; Ni-bearing variant shows additional MC carbide intensity.

(c) Eight-element compact heated to 1000 °C



Multiple sharp low-angle peaks show phase decomposition that contributes to embrittlement.